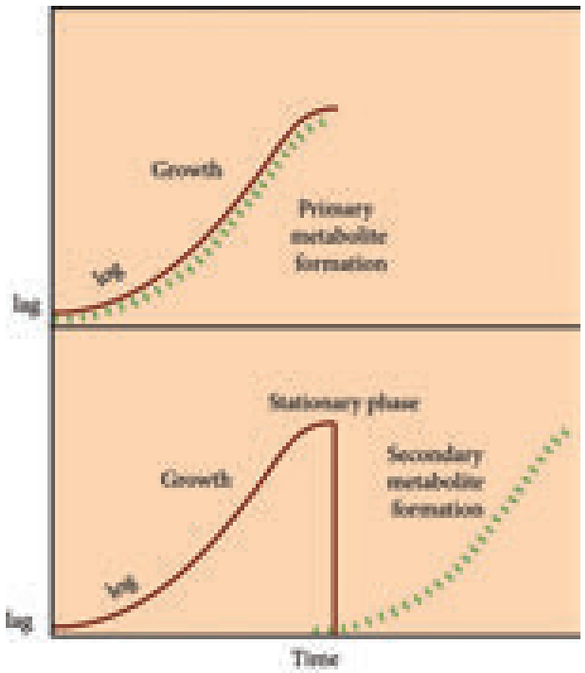


Figure 6.2: Production of primary and secondary metabolites in the growth cycle of microorganism

Some industrially important products are,

- microbial cells (living or dead), microbial biomass and components of microbial cells,
- microbial metabolites,
- intracellular or extracellular enzymes,
- modified compounds that has been microbiologically transformed, and
- recombinant products through the DNA recombinant technology. (Table 6.1 shows some industrially important microorganisms)



The nutritional yeast is called food yeast. The yeast cells are killed during manufacturing, and not alive in the final product. It is used in cooking; it has a cheesy, nutty or savory flavour. Yeast S. *cerevisiae* is used as food yeast. It is a vegan food, available in both fortified (with some vitamins) and unfortified form.

The industrial production of commercial products is carried out by fermentation process. The term fermentation is defined scientifically in a strict sense as a biological process that occurs in the absence oxygen (anaerobic). In industrial sense any process mediated by or involving microorganisms in which a product of economic value is obtained is called fermentation. The term Industrial fermentation also means large scale cultivation of microorganisms even though most of them are aerobic.

Table 6.1: Industrially important microorganisms

Product	Microorganisms	Uses
Vitamin B12	*Streptomyces*	Vitamin supplements
Lactic acid	*Lactobacillus delbrueckii*	Chemical reagents
Citric acid	*Aspergillus niger*	Food preservative
Acetic acid	*Acetobacter*	Vinegar, solvent
Ethanol	*Saccharomyces*	Chemical reagents drinks
Penicillin	*Pencillium chrysogenum*	Antibiotic

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